

Part A — Theory: Regression in Machine Learning

Introduction to Regression

What is Regression in Machine Learning?

Regression is a supervised machine learning technique used to predict numerical values based on patterns found in existing data. In other words, regression helps us estimate a continuous output variable by learning the relationship between input features and the target variable. The goal of a regression model is to identify how changes in one or more variables affect the value of another variable.

Regression is widely used in many real-world applications because many problems involve predicting quantities rather than categories. For example, businesses use regression to forecast sales, hospitals use it to estimate treatment costs, and transportation companies use it to predict travel times. By analyzing historical data, regression models can make predictions that support decision-making and planning.

How is Regression Different from Classification?

Although both regression and classification are supervised learning techniques, they solve different types of problems. Regression predicts continuous numerical values, whereas classification predicts discrete categories or labels.

For example, a regression model may predict the selling price of a house as \$250,000, while a classification model may predict whether an email is "Spam" or "Not Spam." The primary difference lies in the type of output generated by the model.

Real-Life Examples

Regression Example: A car dealership can use regression to predict the selling price of a used car based on factors such as brand, model year, mileage, fuel type, and engine size.

Classification Example: A bank can use classification to determine whether a customer is likely to repay a loan or default on the payment.

Types of Regression

Linear Regression

How It Works

Linear Regression is one of the simplest and most widely used regression algorithms. It assumes that there is a linear relationship between the input variable and the target variable. The model attempts to fit a straight line through the data points by minimizing the prediction errors.

The mathematical equation for simple linear regression is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where:

- Y is the predicted value,
- β_0 is the intercept,
- β_1 is the coefficient,
- X is the input variable,
- ϵ represents the error term.

Real-World Use Case

A company may use linear regression to predict advertising revenue based on the amount spent on marketing campaigns.

Advantages

- Simple and easy to understand.
- Fast to train and implement.
- Works well when relationships are approximately linear.
- Results are highly interpretable.

Limitations

- Assumes a linear relationship.
- Sensitive to outliers.
- Cannot effectively model complex patterns.

Multiple Linear Regression

How It Works

Multiple Linear Regression extends simple linear regression by using multiple input variables to predict a single output variable. Instead of using one feature, it combines several independent variables to improve prediction accuracy.

The equation becomes:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

This approach allows the model to capture the effects of several factors simultaneously.

Real-World Use Case

Real estate companies often use multiple linear regression to predict house prices using factors such as location, number of bedrooms, size, age, and property condition.

Advantages

- Can analyze multiple factors simultaneously.
- Provides interpretable coefficients.
- Usually produces more accurate predictions than simple linear regression.

Limitations

- Assumes linear relationships among variables.
 - May suffer from multicollinearity.
 - Performance decreases when relationships become highly complex.
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Polynomial Regression

How It Works

Polynomial Regression is an extension of linear regression that models nonlinear relationships by adding polynomial terms to the equation. Instead of fitting a straight line, it fits a curved line that can better represent complex patterns.

For example:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$$

Although the equation contains powers of X , the model is still considered a form of linear regression because it remains linear with respect to the coefficients.

Real-World Use Case

Polynomial regression is commonly used in healthcare to model relationships between medication dosage and patient response when the relationship is nonlinear.

Advantages

- Captures nonlinear relationships.
- Provides better flexibility than simple linear models.
- Can improve prediction accuracy for complex datasets.

Limitations

- Easily overfits training data.
 - Computationally more expensive.
 - Difficult to interpret when polynomial degrees become large.
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Regression Metrics

Regression metrics are used to evaluate how accurately a model predicts numerical values.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual values and predicted values. It provides a straightforward interpretation of prediction error.

Formula:

$$\text{MAE} = (1/n) \sum |y - \hat{y}|$$

A lower MAE indicates better performance.

Mean Squared Error (MSE)

MSE calculates the average of squared prediction errors. Because errors are squared, larger errors receive greater penalties.

Formula:

$$\text{MSE} = (1/n) \sum (y - \hat{y})^2$$

Lower values indicate better model performance.

Root Mean Squared Error (RMSE)

RMSE is the square root of MSE. It measures prediction error in the same unit as the target variable, making interpretation easier.

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

A lower RMSE indicates more accurate predictions.

R² (Coefficient of Determination)

R² measures how well the model explains the variation in the target variable. It ranges from 0 to 1, where higher values indicate better performance.

Formula:

$$R^2 = 1 - (SS_{res} / SS_{tot})$$

An R² value close to 1 means the model explains most of the variation in the data.

Comparison of Regression Metrics

Metric	Unit	Sensitive to Large Errors	Meaning
MAE	Same as target	Low	Average absolute prediction error
MSE	Squared units	Very High	Average squared prediction error
RMSE	Same as target	High	Standard deviation of prediction errors
R ²	No unit	No	Percentage of variance explained

Underfitting and Overfitting

Underfitting

Underfitting occurs when a model is too simple to learn the underlying patterns in the data. As a result, it performs poorly on both training and testing datasets.

Overfitting

Overfitting occurs when a model learns the training data too well, including noise and random fluctuations. Although the model performs extremely well on training data, it performs poorly on unseen data.

Causes of Overfitting in Polynomial Regression

Polynomial regression can overfit when the polynomial degree is too high. High-degree polynomials create overly complex curves that fit every training point, including noise and outliers.

Methods to Prevent Overfitting

Several techniques can reduce overfitting:

1. Reduce model complexity.
2. Use cross-validation.

3. Apply regularization methods such as Ridge or Lasso regression.
 4. Increase the amount of training data.
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Real-World Case Study

Predicting House Prices Using Multiple Linear Regression

One of the most widely known applications of regression is house price prediction in the real estate industry. Real estate companies and researchers use multiple linear regression to estimate property values based on various characteristics.

Goal

The goal of the project was to predict housing prices accurately and identify which factors most strongly influence property values.

Data Used

The dataset included variables such as:

- House size,
- Number of bedrooms,
- Number of bathrooms,
- Property age,
- Geographic location,
- Distance to city centers.

Type of Regression Applied

Researchers used Multiple Linear Regression because house prices depend on multiple variables simultaneously.

Results and Insights

The study found that house size, location, and property condition were among the strongest predictors of house prices. Multiple Linear Regression provided interpretable results and helped identify the relative importance of each feature. The model achieved good prediction accuracy and demonstrated how regression techniques can support real-world business decision-making.

References

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